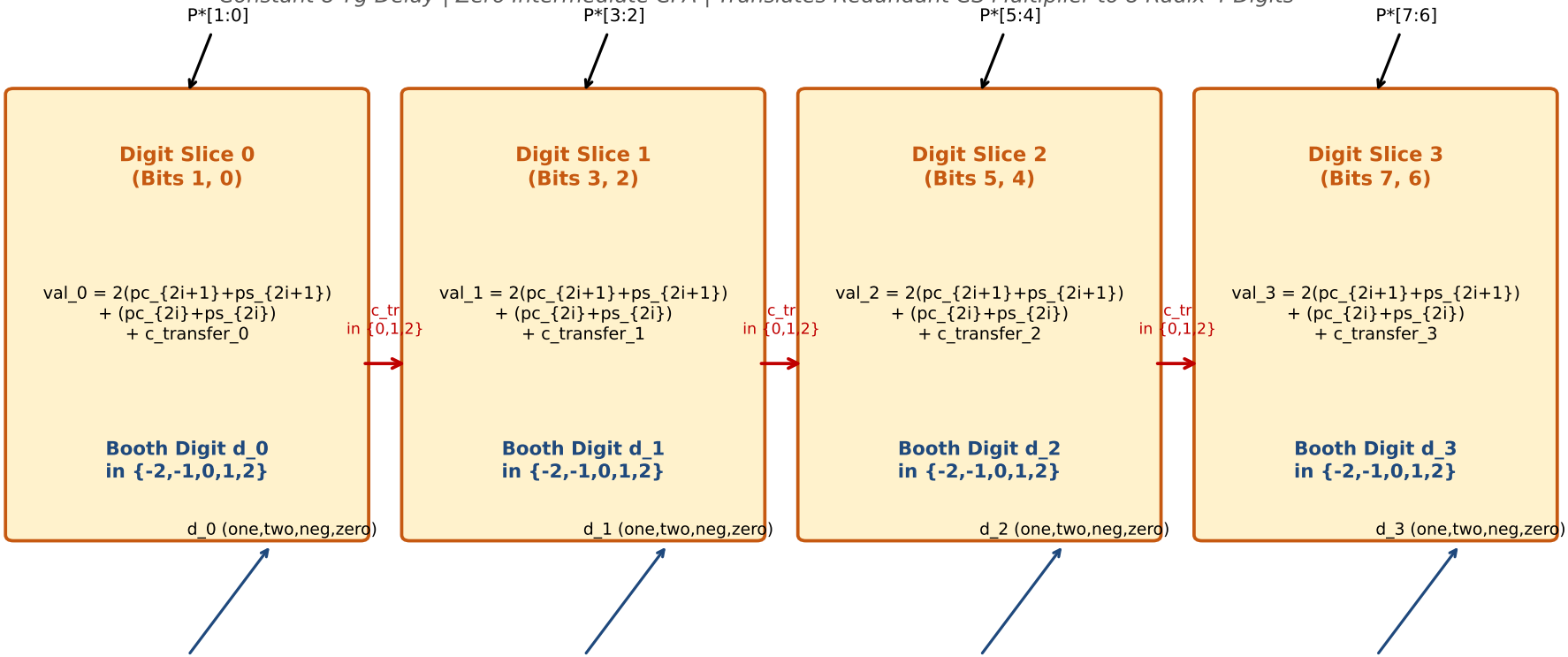
Architecture 5: Direct Carry-Save to Modified Booth (CS-to-MB) Recoder

Constant 8 Tg Delay | Zero Intermediate CPA | Translates Redundant CS Multiplier to 8 Radix-4 Digits



Direct CS-to-MB Recoding Truth Table (val_i in [0, 8] -> Booth Digit d_i & Next Carry):

```
val_i = 0  ->  d_i =  0, c_next = 0 (zero=1)      |      val_i = 5  ->  d_i = +1, c_next = 1 (one=1)
val_i = 1  ->  d_i = +1, c_next = 0 (one=1)       |      val_i = 6  ->  d_i = +2, c_next = 1 (two=1)
val_i = 2  ->  d_i = +2, c_next = 0 (two=1)       |      val_i = 7  ->  d_i = -1, c_next = 2 (neg=1, one=1)
val_i = 3  ->  d_i = -1, c_next = 1 (neg=1, one=1)|      val_i = 8  ->  d_i =  0, c_next = 2 (zero=1)
val_i = 4  ->  d_i =  0, c_next = 1 (zero=1)      |      Transfer carry is bounded strictly to adjacent digits (Delay = 8 Tg!)
```